Supplement to *Gender Differences or Schmifferences: Testing Replication with Open-Source Data*

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Cultural Differences

Research has shown that cultural differences can manifest as personality differences, especially between Western counties (i.e., United States, Canada, Europe) and Asian countries (i.e., Japan, Malaysia, China). In particular, North American countries tend to rate higher in openness to experience, extraversion, and lower in agreeableness (Allik, 2005; Schmitt et al., 2007). Notably, scores tend to converge on the North American and European baseline as socio-economic status increases (Allik, 2005).

To account for potential cultural differences in personality, an independent samples t-test was conducted between North American and Southeast Asian samples. The former was chosen to maintain parity with Weisberg et al. (2011). The latter due to the representativeness of the sample, where participants from Indonesia, Malaysia, and the Philippines accounted for about 64% of the female sample, and 42% of the male sample.

As expected, participants from the Asian sample rated significantly lower in trait openness to experience when compared to the North American sample ($M = 4.44\ vs. M = 4.83, t = -22.51,\ p < .001,\ d = -.30$), with a small effect size. Although the difference was not significant, participants in the Asian sample rated lower on trait agreeableness than participants in the North American sample ($M = 4.57\ vs. M = 4.58,\ t = -.52, p = .601, d = -.01,$). Contrary to expectations, participants in the Asian sample rated significantly higher on trait extraversion than participants in the North American sample ($M = 3.75\ vs. M = 3.71,\ t = 2.39,\ p = .017,\ d = .03,$), with an extremely small effect size. Results are reported in Table S1.

To further assess the sample for cultural differences, independent samples t-tests were conducted on the Asian and North American samples by gender (Tables S2 and S3, respectively). With the samples separated, the change in direction of openness to experience is even more clear, with the Asian females scoring significantly lower on openness to experience than Asian males ($M = 4.40\ vs. M = 4.66, t = -11.83, p < .001,\ d = -.21$). This difference was not significant in the North American sample ($t = .52, p < .602$).

Such differences are also likely to arise with methodological changes between studies (Nosek & Errington, 2020). For instance, participants in the present stated rated personality traits using the Ten-Item Personality Inventory (TIPI; Gosling et al., 2003), whereas Weisberg et al. (2011) used the Big Five Aspect Scales (BFAS), while Lippa and colleagues (2010) reported mean scores from studies using a range of instruments, including the Revised NEO Personality Inventory (NEO-PI-R) and the TIPI. Similarly, Schmitt et al. (2007) used the Big Five Inventory (BFI) while Allik (2005) compared results from both the BFI and NEO-PI-R.

Together, cultural, instrumental, and socio-economic differences between samples are potential explanations for both similarities and differences in the results from the present study and contrast studies.

Table S1

Difference in Average Ten-item Personality Inventory Scores by Asian vs. North American

Scale Sample	n	M	SD	t	df	p	Cohen's d
Openness to Experience				**-22.51**	**13922.6**	**< .001**	**-.30**
Asian	23095	4.44	1.25				
North American	8645	4.83	1.42				
Conscientiousness				**-7.91**	**13901.4**	**< .001**	**-.11**
Asian	23095	4.21	1.41				
North American	8645	4.36	1.61				
Extraversion				**2.39**	**13784.0**	**.017**	**.03**
Asian	23095	3.75	1.19				
North American	8645	3.71	1.37				
Agreeableness				**-.52**	**13145.2**	**.601**	**-.01**
Asian	23095	4.57	1.12				
North American	8645	4.58	1.38				
Emotional Stability				**12.96**	**14348.4**	**< .001**	**.17**
Asian	23095	3.34	1.46				
North American	8645	3.08	1.60				

Table S2

Difference in Average Ten-item Personality Inventory Scores in an Asian Sample by Gender

Scale Gender	n	M	SD	t	df	p	Cohen's d
Openness to Experience				**-11.83**	**5145.0**	**< .001**	**-.21**
Female	19420	4.40	1.24				
Male	3675	4.66	1.25				
Conscientiousness				**.58**	**5122.8**	**.562**	**.01**
Female	19420	4.21	1.41				
Male	3675	4.20	1.42				
Extraversion				**-1.22**	**5130.5**	**.221**	**-.02**
Female	19420	3.75	1.19				
Male	3675	3.77	1.20				
Agreeableness				**4.15**	**5137.5**	**< .001**	**.08**
Female	19420	4.59	1.12				
Male	3675	4.50	1.13				
Emotional Stability				**-17.91**	**4937.1**	**< 001**	**-.34**
Female	19420	3.26	1.43				
Male	3675	3.75	1.55				

Table S3

Difference in Average Ten-item Personality Inventory Scores in a North American Sample by Gender

Scale Gender	n	M	SD	t	df	p	Cohen's d
Openness to Experience				**.52**	**4843.1**	**.602**	**.01**
Female	6000	4.84	1.40				
Male	2645	4.82	1.47				
Conscientiousness				**7.90**	**4937.5**	**< .001**	**.19**
Female	6000	4.46	1.59				
Male	2645	4.16	1.63				
Extraversion				**6.84**	**4996.4**	**< .001**	**.16**
Female	6000	3.78	1.36				
Male	2645	3.56	1.38				
Agreeableness				**11.64**	**4870.4**	**< .001**	**.28**
Female	6000	4.70	1.35				
Male	2645	4.32	1.41				
Emotional Stability				**-19.51**	**4409.2**	**< .001**	**-.48**
Female	6000	2.85	1.48				
Male	2645	3.61	1.74				

Demographic EDA

With respect to gender differences, researchers may be interested in understanding whether pursuit of college majors is evenly distributed by gender (e.g., do males pursue STEM degrees in greater proportion than females?). This section includes figures illustrating proportion of participants by gender and race (Figure S1), proportion of education attainment within the sample by degree and degree by gender (Figure S2), proportion of the sample with college degrees by category and category by gender (Figure S3), and proportion of the with STEM majors by category and category by gender.

Figure S1

Sample Distribution by Gender and Race

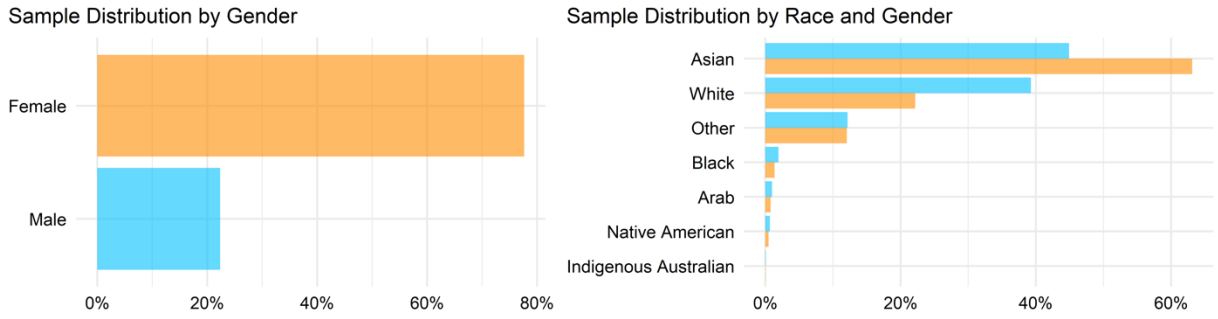


Figure S2

Sample Education Attainment by Level and Gender

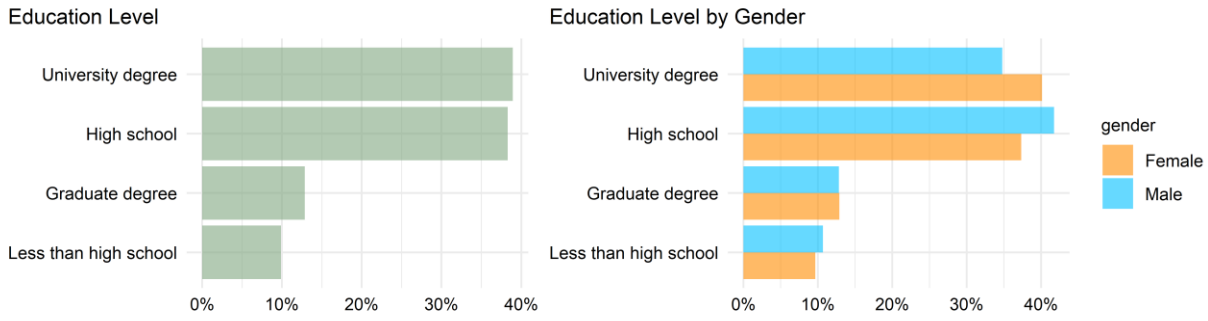


Figure S3

College Majors by Category and Gender

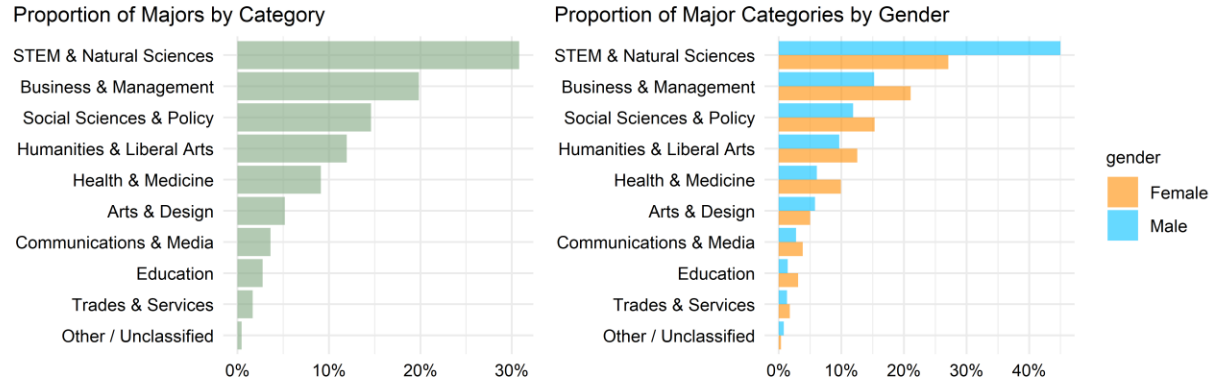
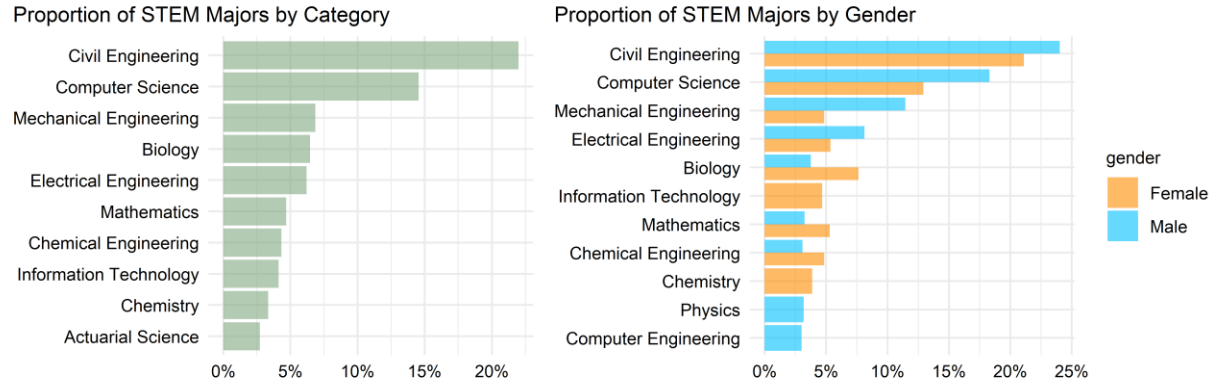


Figure S4

Science, Technology, Engineering, and Math Majors by Category and Gender



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